

Quantum Mechanics

1 Foundations of quantum mechanics

There are some parallels between linear algebra and the mathematical foundation of quantum mechanics. In linear algebra, given a basis $\{\mathbf{e}_i\}$, a n -dimensional complex vector $\mathbf{v}$ can be represented as $(v_1 \dots v_n)$. These vectors live in the vector space $\mathbb{C}^n$. An inner product is defined as $(\mathbf{v}, \mathbf{w}) = v_1^* w_1 + \dots + v_n^* w_n$. There are also linear maps $\mathbb{C}^n \rightarrow \mathbb{C}^n$, represented by a matrix. Now, on the quantum mechanics side, the vectors are states ψ . One of the basis is $\mathbf{x}$ so a state can be represented as $\psi(\mathbf{x}, t)$. They live in $L^2(\mathbb{R}^3)$, the space of complex-valued square-integrable functions. The inner product is

$$(\psi, \phi) = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}, t) \phi(\mathbf{x}, t) d^3x.$$

The linear maps $L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$ are operators.

In classical mechanics, $\mathbf{x}$ and $\dot{\mathbf{x}}$ (or equivalently, $\mathbf{p} = m\dot{\mathbf{x}}$) determine the dynamics of particles in a deterministic way. In quantum mechanics, $\psi(\mathbf{x}, t)$ determines the dynamics of particles in a probabilistic way. ψ is the state of a particle. $\psi(\mathbf{x}, t)$ is the complex coefficient of ψ in the continuous basis of $\mathbf{x}$ at a given time t , that is $\psi(\mathbf{x}, t)$ is ψ in the $\mathbf{x}$ representation and is called the wavefunction. The wavefunction $\psi(\mathbf{x}, t) : \mathbb{R}^3 \rightarrow \mathbb{C}$ (for all $t \in \mathbb{R}$) satisfies mathematical properties dictated by its physical properties.

The probability density for a particle described by the state ψ to sit at $\mathbf{x}$ at a given time t is

$$\rho(\mathbf{x}, t) \propto |\psi(\mathbf{x}, t)|^2.$$

So, $\rho(\mathbf{x}, t)$ is the probability that the particle sits in some small volume dV centred at $\mathbf{x}$. From this probabilistic interpretation, we get the following mathematical properties:

$$\int_{\mathbb{R}^3} \psi^*(\mathbf{x}, t) \psi(\mathbf{x}, t) d^3x = \int_{\mathbb{R}^3} |\psi(\mathbf{x}, t)|^2 d^3x = \mathcal{N} < \infty$$

with $\mathcal{N} \in \mathbb{R}$ and $\mathcal{N} \neq 0$. Since the total probability has to be one, the wavefunction must be normalised to one. The normalised wavefunction being

$$\bar{\psi}(\mathbf{x}, t) = \frac{1}{\sqrt{\mathcal{N}}} \psi(\mathbf{x}, t)$$

and so

$$\int_{\mathbb{R}^3} |\bar{\psi}(\mathbf{x}, t)|^2 d^3x = 1.$$

Often, we just drop the bar notation and normalise it at the end.

Note that if $\tilde{\psi}(\mathbf{x}, t) = e^{i\alpha} \psi(\mathbf{x}, t)$ with $\alpha \in \mathbb{R}$, then $|\tilde{\psi}(\mathbf{x}, t)|^2 = |\psi(\mathbf{x}, t)|^2$, so they are equivalent states. Hence, a state corresponds to rays in the vector space of wavefunctions, $[\psi]$ = the equivalent class of vectors in $L^2(\mathbb{R}^3)$ under the equivalence relation $\psi_1 \sim \psi_2 \iff \psi_1 = e^{i\alpha} \psi_2$.

Definition 1.1 – Hilbert space

The set of all square-integrable functions in $\mathbb{R}^3$ is called Hilbert space $\mathcal{H}$ or $L^2(\mathbb{R}^3)$.

Theorem 1.1

If $\psi_1(\mathbf{x}, t)$ and $\psi_2(\mathbf{x}, t)$ are square-integrable, then

$$\psi(\mathbf{x}, t) = a_1\psi_1(\mathbf{x}, t) + a_2\psi_2(\mathbf{x}, t)$$

with $a_1, a_2 \in \mathbb{C}$ is square-integrable.

Proof. We have

$$\int_{\mathbb{R}^3} |\psi_1(\mathbf{x}, t)|^2 d^3x = \mathcal{N}_1 < \infty \text{ and } \int_{\mathbb{R}^3} |\psi_2(\mathbf{x}, t)|^2 d^3x = \mathcal{N}_2 < \infty.$$

By the triangle inequality, $|z_1 + z_2| \leq |z_1| + |z_2|$ and $(|z_1| - |z_2|)^2 \geq 0$. If we take $z_1 = a_1\psi_1(\mathbf{x}, t)$ and $z_2 = a_2\psi_2(\mathbf{x}, t)$, we get

$$\begin{aligned} \int_{\mathbb{R}^3} |\psi|^2 d^3x &= \int_{\mathbb{R}^3} |a_1\psi_1 + a_2\psi_2|^2 d^3x \\ &\leq \int_{\mathbb{R}^3} (|a_1\psi_1| + |a_2\psi_2|)^2 d^3x \\ &= \int_{\mathbb{R}^3} (|a_1\psi_1|^2 + |a_2\psi_2|^2 + 2|a_1\psi_1||a_2\psi_2|) d^3x \\ &\leq \int_{\mathbb{R}^3} (2|a_1\psi_1|^2 + 2|a_2\psi_2|^2) d^3x \\ &= 2|a_1|^2 \mathcal{N}_1 + 2|a_2|^2 \mathcal{N}_2 < \infty. \end{aligned}$$

□

Definition 1.2 – Inner product on $\mathcal{H}$

$$(\psi, \phi) = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}, t) \phi(\mathbf{x}, t) d^3x$$

Theorem 1.2

If $\psi, \phi \in \mathcal{H}$, then their inner product is guaranteed to exist.

Proof. Similar to the last proof, we have

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x}, t)|^2 d^3x = \mathcal{N}_1 < \infty \text{ and } \int_{\mathbb{R}^3} |\phi(\mathbf{x}, t)|^2 d^3x = \mathcal{N}_2 < \infty.$$

By the Cauchy-Schwarz inequality,

$$\begin{aligned} |(\psi, \phi)| &= \left| \int_{\mathbb{R}^3} \psi^* \phi d^3x \right| \\ &\leq \sqrt{\int_{\mathbb{R}^3} |\psi|^2 d^3x \int_{\mathbb{R}^3} |\phi|^2 d^3x} \\ &= \sqrt{\mathcal{N}_1 \mathcal{N}_2} < \infty. \end{aligned}$$

□

(i) $(\psi, \phi) = (\phi, \psi)^*$.

(ii) antilinear in the first entry and linear in the second entry, that is for $a_1, a_2 \in \mathbb{C}$, we have

$$(a_1\psi_1 + a_2\psi_2, \phi) = a_1^*(\psi_1, \phi) + a_2^*(\psi_2, \phi) \text{ and } (\psi, a_1\phi_1 + a_2\phi_2) = a_1(\psi, \phi_1) + a_2(\psi, \phi_2).$$

(iii) $(\psi, \psi) \geq 0$ and it is zero iff $\psi = 0$, which is not an admissible physical state.

Definition 1.3 – Norm

The norm of a wavefunction is a real function

$$\|\psi\| = \sqrt{(\psi, \psi)}.$$

We say $\psi(\mathbf{x}, t)$ is normalised if $\|\psi\| = 1$. Two wavefunctions $\psi, \phi \in \mathcal{H}$ are orthogonal if $(\psi, \phi) = 0$.

Definition 1.4 – Orthonormal set

A set of wavefunctions $\{\psi_n\}$ is orthonormal if

$$(\psi_m, \psi_n) = \delta_{mn}.$$

Definition 1.5 – Complete set

A set of wavefunctions $\{\psi_n\}$ is complete if any $\phi \in \mathcal{H}$ can be written as a linear combination of them, that is

$$\phi = \sum_{n=0}^{\infty} c_n \psi_n$$

for some $\{c_n\} \subseteq \mathbb{C}$.

Lemma 1.1

If $\{\psi_n\}$ forms a complete and orthonormal basis of $\mathcal{H}$, then

$$c_n = (\psi_n, \phi).$$

Proof. We use Fourier's trick,

$$\begin{aligned} (\psi_n, \phi) &= \left(\psi_n, \sum_{m=0}^{\infty} c_m \psi_m \right) \\ &= \sum_{m=0}^{\infty} c_m (\psi_n, \psi_m) \\ &= \sum_{m=0}^{\infty} c_m \delta_{nm} = c_n \end{aligned}$$

□

The first postulate of quantum mechanics is Born's rule $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$. The second postulate is the

Definition 1.6 – Time-dependent Schrödinger equation

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + U(\mathbf{x}) \psi(\mathbf{x}, t)$$

where $U(\mathbf{x}) \in \mathbb{R}$ is the potential.

Notice the first derivative in time, so once $\psi(\mathbf{x}, t_0)$ is known, we can get $\psi(\mathbf{x}, t)$. Also, there is an asymmetry between t and $\mathbf{x}$, so this a non-relativistic equation.

Proposition 1.1

If ψ obeys the TDSE, then the normalisation is constant.

Proof. Let $\int_{\mathbb{R}^3} |\psi|^2 d^3x = \mathcal{N}$.

$$\begin{aligned}
 \frac{d\mathcal{N}}{dt} &= \frac{d}{dt} \int_{\mathbb{R}^3} |\psi|^2 d^3x \\
 &= \int_{\mathbb{R}^3} \frac{\partial}{\partial t} (\psi^* \psi) d^3x \\
 &= \int_{\mathbb{R}^3} \left[\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right] d^3x \\
 &= \int_{\mathbb{R}^3} \left[\psi^* \left(\frac{i\hbar}{2m} \nabla^2 \psi - \frac{iU}{\hbar} \psi \right) + \psi \left(-\frac{i\hbar}{2m} \nabla^2 \psi^* + \frac{iU}{\hbar} \psi^* \right) \right] d^3x \\
 &= \int_{\mathbb{R}^3} \nabla \cdot \left[\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right] d^3x = 0
 \end{aligned}$$

since $|\psi|, |\psi^*| \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$. We have taken the complex conjugate of the TDSE to get $\partial \psi^* / \partial t$. $\square$

We see from above that

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left[-\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right] = 0$$

so we can define the

Definition 1.7 – Probability current

$$\mathbf{j} = -\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

This is the conservation of probability in quantum mechanics.

How do we extract information from a state? An observable is any property of the particle described by ψ that can be measured. In quantum mechanics, an observable corresponds to an operator acting on ψ and a measurement is the expectation value of an operator.

From the probabilistic interpretation, if we want to measure the position of a particle,

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\psi|^2 dx = \int_{-\infty}^{\infty} \psi^* x \psi dx.$$

The expectation value of an observable is just the mean of an infinite series of measurements performed on

particles with the same state ψ . What about momentum?

$$\begin{aligned}
\langle p \rangle &= m \frac{d\langle x \rangle}{dt} \\
&= m \frac{d}{dt} \int_{-\infty}^{\infty} \psi^* x \psi dx \\
&= m \int_{-\infty}^{\infty} x \frac{\partial}{\partial t} (\psi^* \psi) dx \\
&= \frac{i\hbar m}{2m} \int_{-\infty}^{\infty} x \frac{\partial}{\partial x} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) dx \quad \text{by the TDSE} \\
&= \frac{i\hbar}{2} \left[\left[x \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) dx \right] \\
&= -i\hbar \int_{-\infty}^{\infty} \psi^* \frac{\partial \psi}{\partial x} dx \quad \text{since the boundary terms vanish} \\
&= \int_{-\infty}^{\infty} \psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \psi dx
\end{aligned}$$

We see that the position operator $\hat{x}$ is x and the momentum operator $\hat{p}$ is $-i\hbar \partial/\partial x$.

Linear maps from $\mathbb{C}^n$ to $\mathbb{C}^n$ are given by matrices. In Hilbert space, linear maps are given by operators.

Definition 1.8 – Operator

An operator $\hat{O}$ is a linear map $\mathcal{H} \rightarrow \mathcal{H}$ such that

$$\hat{O}(a_1\psi_1 + a_2\psi_2) = a_1\hat{O}(\psi_1) + a_2\hat{O}(\psi_2)$$

with $a_1, a_2 \in \mathbb{C}$ and $\psi_1, \psi_2 \in \mathcal{H}$.

Some examples include finite differential operators

$$\sum_{n=0}^N a_n(x) \frac{\partial^n}{\partial x^n}$$

where $a_n(x)$ are polynomials, the translation operator $\hat{S}_a : \psi(x) \mapsto \psi(x - a)$ with $a \in \mathbb{R}$, and the parity operator $\hat{P} : \psi(x) \mapsto \psi(-x)$.

Definition 1.9 – Hermitian conjugate

The Hermitian conjugate $\hat{O}^\dagger$ of an operator $\hat{O}$ is the operator such that

$$(\hat{O}^\dagger \psi_1, \psi_2) = (\psi_1, \hat{O} \psi_2)$$

for all $\psi_1, \psi_2 \in \mathcal{H}$.

We can verify that $(a_1\hat{A}_1 + a_2\hat{A}_2)^\dagger = a_1^*\hat{A}_1^\dagger + a_2^*\hat{A}_2^\dagger$ where $a_1, a_2 \in \mathbb{C}$ and $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$.

Definition 1.10 – Hermitian operator

An operator $\hat{O}$ is Hermitian if $\hat{O} = \hat{O}^\dagger$ or equivalently

$$(\hat{O} \psi_1, \psi_2) = (\psi_1, \hat{O} \psi_2)$$

for all $\psi_1, \psi_2 \in \mathcal{H}$.

All physical quantities in quantum mechanics are represented by Hermitian operators. For example, $\hat{x}$ is

Hermitian since

$$(\hat{x}\psi_1, \psi_2) = \int_{-\infty}^{\infty} \psi_1^* x \psi_2 dx = (\psi_1, \hat{x}\psi_2).$$

For the momentum operator $\hat{p} : \psi \mapsto -i\hbar \partial\psi/\partial x$,

$$\begin{aligned} (\hat{p}\psi_1, \psi_2) &= \int_{-\infty}^{\infty} \left(-i\hbar \frac{\partial\psi_1}{\partial x}\right)^* \psi_2 dx \\ &= i\hbar \int_{-\infty}^{\infty} \frac{\partial\psi_1^*}{\partial x} \psi_2 dx \\ &= i\hbar [\psi_1^* \psi_2]_{-\infty}^{\infty} - i\hbar \int_{-\infty}^{\infty} \psi_1^* \frac{\partial\psi_2}{\partial x} dx \\ &= \int_{-\infty}^{\infty} \psi_1^* \left(-i\hbar \frac{\partial\psi_2}{\partial x}\right) dx = (\psi_1, \hat{p}\psi_2). \end{aligned}$$

The kinetic energy operator

$$\hat{T} : \psi \mapsto \frac{\hat{p}^2}{2m} \psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2},$$

the potential energy operator $\hat{U} : \psi \mapsto U\psi$, and the total energy operator

$$\hat{H} : \psi \mapsto (\hat{T} + \hat{U})\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + U\psi$$

are all Hermitian.

Theorem 1.3

The eigenvalues of Hermitian operators are real.

Proof. Let $\hat{A}$ be a Hermitian operator with eigenvalue $a \in \mathbb{C}$ and eigenfunction $\psi \in \mathcal{H}$ with $\|\psi\| = 1$.

$$(\psi, \hat{A}\psi) = (\psi, a\psi) = a(\psi, \psi) = a\|\psi\|^2 = a$$

But $\hat{A}$ is Hermitian, so

$$(\psi, \hat{A}\psi) = (\hat{A}\psi, \psi) = (a\psi, \psi) = a^*(\psi, \psi) = a^*\|\psi\|^2 = a^*$$

Hence, $a = a^*$. □

Theorem 1.4

If $\hat{A}$ is a Hermitian operator and ψ_1, ψ_2 are normalised eigenfunctions of $\hat{A}$ with eigenvalues a_1, a_2 respectively with $a_1 \neq a_2$, then ψ_1 and ψ_2 are orthogonal.

Proof. We have $\hat{A}\psi_1 = a_1\psi_1$ and $\hat{A}\psi_2 = a_2\psi_2$ with $a_1, a_2 \in \mathbb{R}$.

$$a_1(\psi_1, \psi_2) = a_1^*(\psi_1, \psi_2) = (a_1\psi_1, \psi_2) = (\hat{A}\psi_1, \psi_2) = (\psi_1, \hat{A}\psi_2) = (\psi_1, a_2\psi_2) = a_2(\psi_1, \psi_2),$$

so $(a_1 - a_2)(\psi_1, \psi_2) = 0$. Hence, $(\psi_1, \psi_2) = 0$ since $a_1 \neq a_2$. □

Theorem 1.5

The discrete (or continuous) set of eigenfunctions of any Hermitian operator forms a complete orthonormal basis of $\mathcal{H}$.

That is, if $\{\psi_i\}$ is the set of eigenfunctions, then any $\psi \in \mathcal{H}$ can be expressed as

$$\psi(\mathbf{x}, t) = \sum_{i=1}^N a_i \psi_i(\mathbf{x}, t)$$

where $a_i \in \mathbb{C}$ and N possibly infinite. For a continuous set, then we have

$$\psi(\mathbf{x}, t) = \int_{\Delta\lambda} a(\lambda) \psi_\lambda(\mathbf{x}, t) d\lambda$$

where $\Delta\lambda \subseteq \mathbb{R}$.

So far, we know that quantum observables correspond to Hermitian operators. To make this concrete, we have the Copenhagen interpretation:

- (i) The possible outcomes of a measurement of an observable O are the eigenvalues of $\hat{O}$.
- (ii) If $\hat{O}$ has a discrete set of normalised eigenfunctions $\{\psi_i\}$ with distinct eigenvalues $\{\lambda_i\}$, then the measurement of O on a particle with state ψ has probability

$$\mathbb{P}(O = \lambda_i) = |a_i|^2 = |(\psi_i, \psi)|^2.$$

- (iii) If $\{\psi_i\}$ are the orthonormal eigenfunctions of $\hat{O}$ and $\{\psi_i\}_{i \in I}$ are a complete set of orthonormal eigenfunctions with the same eigenvalue λ , then

$$\mathbb{P}(O = \lambda) = \sum_{i \in I} |a_i|^2 = \sum_{i \in I} |(\psi_i, \psi)|^2.$$

- (iv) The projection postulate, if O is measured on ψ at time t and the outcome of the measurement is λ_i , then the wavefunction of ψ instantaneously after the measurement becomes ψ_i , the eigenfunction of $\hat{O}$ with eigenvalue λ_i . If $\hat{O}$ has degenerate eigenfunctions with the same eigenvalue λ , then ψ collapses to $\sum_{i \in I} a_i \psi_i$.

As a sanity check, the sum of all the probabilities is one,

$$\sum_{i=1}^N |a_i|^2 = \sum_{i=1}^N (a_i \psi_i, a_i \psi_i) = \sum_{i,j=1}^N (a_i \psi_i, a_j \psi_j) = (\psi, \psi) = 1$$

The physics implication of the projection postulate is that if an observable is measured twice, the outcome of the second measurement (if $\Delta t \rightarrow 0$) is the same as the first with probability one.

Let $\psi = \sum_{i=1}^N a_i \psi_i = \sum_{i=1}^N (\psi_i, \psi) \psi_i$, the projection operator is

$$\hat{P}_i = \psi \mapsto (\psi_i, \psi) \psi_i.$$

The expectation value of O measured on ψ is

$$\begin{aligned} \langle O \rangle_\psi &= \sum_{i=1}^N \lambda_i \mathbb{P}(O = \lambda_i) \\ &= \sum_{i=1}^N \lambda_i |a_i|^2 = \sum_{i=1}^N \lambda_i |(\psi_i, \psi)|^2 \\ &= \left(\sum_{i=1}^N (\psi_i, \psi) \psi_i, \sum_{j=1}^N \lambda_j (\psi_j, \psi) \psi_j \right) \\ &= (\psi, \hat{O} \psi) = \int_{\mathbb{R}^3} \psi^* \hat{O} \psi d\mathbf{x}. \end{aligned}$$

This is linear, $\langle a\hat{A} + b\hat{B} \rangle_\psi = a\langle \hat{A} \rangle_\psi + b\langle \hat{B} \rangle_\psi$ with $a, b \in \mathbb{C}$.

We can generalise the projection postulate beyond ψ_i with

Theorem 1.6 – Born's rule

If ϕ is the state that gives the desired outcome of a measurement of a particle described by state ψ , then the probability that a measurement on ψ gives such outcome is given by

$$|(\phi, \psi)|^2 = \left| \int_{\mathbb{R}^3} \phi^* \psi d\mathbf{x} \right|^2.$$

Let us rewrite the TDSE in 1D,

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + U\psi = \hat{H}\psi$$

where $\hat{H}$ is the Hamiltonian operator. Here we assume U is independent of t , that is $U = U(x)$. We do separation of variables and try $\psi(x, t) = T(t)\chi(x)$,

$$i\hbar \frac{\partial T(t)}{\partial t} \chi(x) = T(t) \hat{H} \chi(x) \implies \frac{1}{T(t)} i\hbar \frac{\partial T(t)}{\partial t} = \frac{\hat{H} \chi(x)}{\chi(x)}.$$

Since the left only depends on t while the right only depends on x , both sides must be constant. Calling this constant E ,

$$\frac{1}{T(t)} i\hbar \frac{\partial T(t)}{\partial t} = E \implies T(t) = e^{-iEt/\hbar}.$$

The $\chi(x)$ side gives the

Theorem 1.7 – Time-independent Schrödinger equation

$$\hat{H}\chi = -\frac{\hbar^2}{2m} \frac{\partial^2 \chi}{\partial x^2} + U\chi = E\chi$$

This is an eigenvalue equation for the $\hat{H}$ operator. The eigenvalues of $\hat{H}$ are all the possible outcomes of measuring energy on the state ψ .

We found a particular solution of the TDSE

$$\psi(x, t) = e^{-iEt/\hbar} \chi(x)$$

where E is the eigenvalue associated with the eigenfunction. These particular solutions are called stationary states. They are called stationary as the probability density does not change with time, $\rho(x, t) = |e^{-iEt/\hbar} \chi(x)|^2 = |\chi(x)|^2$.

Every solution of the TDSE can be written as a linear combination of stationary states. For a system with a discrete set of eigenvalues of $\hat{H}$, we can write the solution as

$$\psi(x, t) = \sum_n a_n \chi_n(x) e^{-iE_n t/\hbar}.$$

For a system that has a continuous set of eigenvalues of $\hat{H}$, we have

$$\psi(x, t) = \int_{\Delta\alpha} A(\alpha) \chi_\alpha(x) e^{-iE_\alpha t/\hbar} d\alpha$$

where $A \in \mathbb{C}$ and $\alpha \in \mathbb{R}$. By the Copenhagen interpretation, $|a_n|^2$ or $|A(\alpha)|^2 d\alpha$ is the probability of measuring the energy of a particle to be E_n .

Example 1.1

Imagine a system with only two energy eigenvalues $E_1 \neq E_2$. Then, we can write

$$\psi(x, t) = a_1 \chi_1(x) e^{-iE_1 t/\hbar} + a_2 \chi_2(x) e^{-iE_2 t/\hbar}.$$

At $t = 0$, we have $\psi(x, 0) = a_1 \chi_1(x) + a_2 \chi_2(x)$. If either a_1 or a_2 is zero, then ψ is a stationary state. If $a_1, a_2 \neq 0$, then

$$\begin{aligned} \rho(x, t) &= |\psi(x, t)|^2 \\ &= (a_1 \chi_1 e^{-iE_1 t/\hbar} + a_2 \chi_2 e^{-iE_2 t/\hbar})^* (a_1 \chi_1 e^{-iE_1 t/\hbar} + a_2 \chi_2 e^{-iE_2 t/\hbar}) \\ &= a_1^2 |\chi_1|^2 + a_2^2 |\chi_2|^2 + a_1 a_2 \chi_1 \chi_2 (e^{i(E_1 - E_2)t/\hbar} + e^{i(E_2 - E_1)t/\hbar}) \\ &= a_1^2 |\chi_1|^2 + a_2^2 |\chi_2|^2 + 2a_1 a_2 \chi_1 \chi_2 \cos((E_1 - E_2)t/\hbar). \end{aligned}$$

Note that we have chosen the energy eigenfunctions to be real (see example sheet for more details).

2 1D solutions to the Schrödinger equation

2.1 Infinite potential well

Consider

$$U(x) = \begin{cases} 0 & \text{if } |x| \leq a \\ \infty & \text{if } |x| > a \end{cases}$$

with $a > 0$. For $|x| > a$, $\chi(x) = 0$ since otherwise $U\chi = \infty$. For $|x| \leq a$, we look for solutions to

$$-\frac{\hbar^2}{2m} \chi'' = E\chi \implies \chi'' + \frac{2mE}{\hbar^2} \chi = 0.$$

Letting $k = \sqrt{2mE}/\hbar$, the solution is $\chi(x) = A \sin(kx) + B \cos(kx)$. We want to impose the boundary conditions that χ is continuous, so $\chi(a) = \chi(-a) = 0$, which gives $A \sin(ka) = B \cos(ka) = 0$. There are two options.

- (i) $A = 0$ and $\cos(ka) = 0$, which gives $k_n = n\pi/(2a)$ where n is an odd positive integer. So, $\chi_n(x) = B \cos(k_n x)$ which are the even solutions.
- (ii) $B = 0$ and $\sin(ka) = 0$, which gives $k_n = n\pi/(2a)$ where n is an even positive integer. So, $\chi_n(x) = A \sin(k_n x)$ which are the odd solutions.

The eigenvalues of $\hat{H}$ are then

$$E_n = \frac{\hbar^2}{2m} k_n^2 = \frac{\hbar^2 \pi^2}{8ma^2} n^2.$$

We can normalise χ_n to get $A = B = \sqrt{1/a}$. So, the normalised eigenfunctions of $\hat{H}$ are

$$\chi_n(x) = \begin{cases} \sqrt{1/a} \cos(n\pi x/2a) & \text{for } n = 1, 3, 5, \dots \\ \sqrt{1/a} \sin(n\pi x/2a) & \text{for } n = 2, 4, 6, \dots \end{cases}$$

We see that the ground state has nonzero energy, contradictory to classical mechanics. Also, as $n \rightarrow \infty$, the probability density approaches a constant, demonstrating the correspondence principle.

There are some conditions $\chi(x)$ must satisfy. If $U(x)$ is finite, then $\chi(x)$ must be twice differentiable and the second derivative must exist and be finite. If $U(x)$ is infinite in some regions, then $\chi(x)$ must be continuous but $\chi'(x)$ does not have to be. In fact, $\chi'(x)$ is discontinuous when the potential jumps off to infinity at a point.

Proposition 2.1

If a quantum system has non-degenerate eigenstates, that is $E_i \neq E_j$ if $i \neq j$, then if $U(x) = U(-x)$, the eigenfunctions of $\hat{H}$ have to be either odd or even.

Proof. If $U(x) = U(-x)$, then the TISE is invariant under $x \mapsto -x$. If $\chi(x)$ is a solution with eigenvalue E , then $\chi(-x)$ is also a solution with the same eigenvalue. So, $\chi(x)$ is equivalent to $\chi(-x)$, that is $\chi(x) = \lambda\chi(-x)$ with $\lambda \in \mathbb{C}$ and $|\lambda| = 1$. $\chi(x) = \chi(-(-x)) = \lambda\chi(-x) = \lambda^2\chi(x)$. Hence, $\lambda = \pm 1$ and $\chi(x) = \pm\chi(-x)$. $\square$

2.2 Finite potential well

Consider

$$U(x) = \begin{cases} 0 & \text{if } |x| \leq a \\ U_0 & \text{if } |x| > a \end{cases}$$

with $a, U_0 > 0$. Here, $E > 0$, and we only look at bound states $E < U_0$. Since $U(x)$ is symmetric, we only look at even solutions here. See example sheets for the odd solutions. The TISE is

$$\begin{cases} -\frac{\hbar^2}{2m}\chi''(x) = E\chi(x) & \text{if } |x| \leq a \\ -\frac{\hbar^2}{2m}\chi''(x) = (E - U_0)\chi(x) & \text{if } |x| > a. \end{cases}$$

The first equation simplifies to $\chi''(x) + k^2\chi(x) = 0$ where $k = \sqrt{2mE}/\hbar$ and the solution is $\chi(x) = A \sin(kx) + B \cos(kx)$. Since we only consider even solutions now, we have $\chi(x) = B \cos(kx)$.

The second equation simplifies to $\chi''(x) - \bar{k}^2\chi(x) = 0$ where $\bar{k} = \sqrt{2m(U_0 - E)}/\hbar$ and the solution is $\chi(x) = Ce^{\bar{k}x} + De^{-\bar{k}x}$. If $C \neq 0$, then $\chi(x)$ blows up as $x \rightarrow \infty$, so $C = 0$ when $x > a$. Similarly, $D = 0$ when $x < -a$. Hence,

$$\begin{cases} Ce^{\bar{k}x} & \text{if } x < -a \\ B \cos(kx) & \text{if } -a \leq x \leq a \\ De^{-\bar{k}x} & \text{if } x > a. \end{cases}$$

Imposing continuity of $\chi(x)$ and $\chi'(x)$ at $x = \pm a$, we have

$$\begin{cases} Ce^{-\bar{k}a} = B \cos(ka) \\ -\bar{k}Ce^{-\bar{k}a} = -kB \sin(ka) \end{cases}$$

at $x = -a$. Taking the ratio of the equations, $k \tan(ka) = \bar{k}$, and from the definition of k and $\bar{k}$, we have $k^2 + \bar{k}^2 = 2mU_0/\hbar^2$. Letting $\xi = ka$ and $\eta = \bar{k}a$, we get

$$\begin{cases} \xi \tan \xi = \eta \\ \xi^2 + \eta^2 = r_0^2 \end{cases}$$

where $r_0^2 = 2mU_0a^2/\hbar^2$. The eigenvalues of $\hat{H}$ corresponds to the points of intersection, which there are a finite number, with

$$E_n = \frac{\hbar^2}{2ma^2}\xi_n^2 \text{ with } n = 1, 2, \dots, p.$$

As $U_0 \rightarrow \infty$, the solutions tend to that of the infinite potential well. To get the full solution, we can solve for C in terms of B and use normalisation.

2.3 Harmonic oscillator

We have $U(x) = kx^2/2$ where $k \in \mathbb{R}^+$ is the elastic constant. Setting $\omega = \sqrt{k/m}$, the TISE is

$$-\frac{\hbar^2}{2m}\chi''(x) + \frac{1}{2}m\omega^2 x^2 \chi(x) = E\chi(x).$$

We know that it has discrete eigenvalues and the eigenfunctions are odd or even. We perform a change of variables $\xi^2 = m\omega x^2/\hbar$ and $\varepsilon = 2E/(\hbar\omega)$ and we get

$$-\frac{d^2}{d\xi^2}\chi(\xi) + \xi^2\chi(\xi) = \varepsilon\chi(\xi).$$

We solve it by starting from a particular solution with $\varepsilon = 1$ and $E_0 = \hbar\omega/2$. The ansatz $\chi_0(\xi) = e^{-\xi^2/2}$ works, so we found one eigenvalue $E_0 = \hbar\omega/2$ with eigenfunction $\chi_0(x) = Ae^{-m\omega x^2/(2\hbar)}$. To find the other eigenfunctions of $\hat{H}$, we use the general form $\chi(\xi) = f(\xi)e^{-\xi^2/2}$, and plugging this in,

$$-\frac{d^2 f}{d\xi^2} + 2\xi \frac{df}{d\xi} + (1 - \varepsilon)f = 0.$$

To solve this, we use a power series for the form

$$f(\xi) = \sum_{n=0}^{\infty} a_n \xi^n$$

with $a_n \in \mathbb{R}$. We get

$$\sum_{n=0}^{\infty} [(n+1)(n+2)a_{n+2} - 2na_n + (\varepsilon - 1)a_n]\xi^n = 0$$

so

$$a_{n+2} = \frac{2n - \varepsilon + 1}{(n+1)(n+2)} a_n.$$

From the parity of the wavefunction, either $a_n = 0$ for all odd n if f is even, or $a_n = 0$ for all even n if f is odd.

Proposition 2.2

If a series solution $f(\xi) = \sum_{n=0}^{\infty} a_n \xi^n$ for the harmonic oscillator does not terminate, then it cannot be normalised.

Proof. We look at the asymptotic behaviour of the series, $a_{n+2}/a_n \rightarrow 2/n$ as $n \rightarrow \infty$. This is the same as $g(\xi) = e^{\xi^2} = \sum_{n=0}^{\infty} \xi^{2n}/n!$. So, $\chi(\xi) \sim e^{\xi^2} e^{-\xi^2/2} = e^{\xi^2/2}$ and it cannot be normalised. $\square$

Given that the series must terminate, there exists N such that $a_{N+2} = 0$ with $a_N \neq 0$. So,

$$a_{N+2} = \frac{2N - \varepsilon + 1}{(N+1)(N+2)} a_N = 0$$

and $2N - \varepsilon + 1 = 0$. This gives

$$E_N = (N + 1/2)\hbar\omega$$

with eigenfunctions

$$\chi_N(\xi) = f_N(\xi)e^{-\xi^2/2}$$

where $f_N(\xi)$ are called the Hermite polynomials, either defined by the recursive relation above or with

$$f_N(\xi) = (-1)^N e^{\xi^2} \frac{d^N}{d\xi^N} (e^{-\xi^2}).$$

2.4 The free particle

The TISE is

$$-\frac{\hbar^2}{2m}\chi''(x) = E\chi(x)$$

since $U(x) = 0$. Letting $k = \sqrt{2mE}/\hbar$, we get $\chi''(x) + k^2\chi(x) = 0$ with solution $\chi(x) = Ce^{ikx}$. The energy values are continuous instead of discrete, with

$$E_k = \frac{\hbar^2 k^2}{2m}$$

where $k \in \mathbb{R}$. The time-dependent solution is

$$\psi_k(x, t) = e^{i(kx - \hbar k^2 t / (2m))}$$

but this is not square-integrable. We can deal with these unbounded states in two ways.

- (i) Build a linear superposition of χ_k that is normalisable.
- (ii) Ignore the problem and change the interpretation.

We go with the first option. Write

$$\psi(x, t) = \int_{-\infty}^{\infty} A(k) \psi_k(x, t) dk$$

with $A(k) \in \mathbb{R}$. We can choose $A(k)$ freely but one possible option is the Gaussian wavepacket

$$A(k) = \exp\left[-\frac{\sigma}{2}(k - k_0)^2\right]$$

with $\sigma \in \mathbb{R}^+$ and $k_0 \in \mathbb{R}$.

$$\psi_{\text{GP}}(x, t) = \int_{-\infty}^{\infty} \exp[F(k)] dk$$

where

$$\begin{aligned} F(k) &= -\frac{\sigma}{2}(k - k_0)^2 + ikx - i\frac{\hbar k^2}{2m}t \\ &= -\frac{1}{2} \underbrace{\left(\sigma + \frac{i\hbar t}{m}\right)}_{\alpha} k^2 + \underbrace{(k_0\sigma + ix)}_{\beta} k - \underbrace{\frac{\sigma}{2}k_0^2}_{\delta} \\ &= -\frac{\alpha}{2} \left(k - \frac{\beta}{\alpha}\right)^2 + \frac{\beta^2}{2\alpha} + \delta \end{aligned}$$

so

$$\psi_{\text{GP}}(x, t) = \exp\left[\frac{\beta^2}{2\alpha} + \delta\right] \int_{-\infty}^{\infty} \exp\left[-\frac{\alpha}{2} \left(k - \frac{\beta}{\alpha}\right)^2\right] dk.$$

Let $\tilde{k} = k - \beta/\alpha$ and $\text{Im}(\tilde{k}) = \text{Im}(\beta/\alpha) = \nu$, we have

$$\psi_{\text{GP}}(x, t) = \exp\left[\frac{\beta^2}{2\alpha} + \delta\right] \int_{-\infty - i\nu}^{\infty - i\nu} \exp\left[-\frac{\alpha}{2}\tilde{k}^2\right] d\tilde{k} = \exp\left[\frac{\beta^2}{2\alpha} + \delta\right] \sqrt{\frac{2\pi}{\alpha}}.$$

Since β^2 has a $-x^2$ term, the wavefunction is normalisable and we get

$$|\psi_{\text{GP}}(x, t)|^2 = \sqrt{\frac{\sigma}{\pi(\sigma^2 + \hbar^2 t^2 / m^2)}} \exp\left[-\frac{\sigma(x - \hbar k_0 t / m)^2}{\sigma^2 + \hbar^2 t^2 / m^2}\right].$$

The expected value of position is $\langle x \rangle = \hbar k_0 t / m$ so the particle is moving with velocity $\hbar k_0 / m$. The error in

position is

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{1}{2} \left(\sigma + \frac{\hbar^2 t^2}{m^2 \sigma} \right)}$$

and we see that the error increases with time. For momentum, $\langle \hat{p} \rangle = \hbar k_0$ and

$$\Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2} = \frac{\hbar}{\sqrt{2\sigma}}$$

for all time. At $t = 0$, $\Delta x \Delta p = \hbar/2$, and as time increases, the uncertainty increases.

What about the second option? It provides a faster way to do these calculations. Just ignore the normalisation problem and take $\chi_k(x) = e^{ikx}$ to be an eigenfunction of $\hat{H}$ so

$$\psi_k(x, t) = A e^{ikx} e^{-i\hbar k^2 t / (2m)}$$

where $A \in \mathbb{C}$. We adopt the beam interpretation, instead of $\psi_k(x, t)$ describing a single particle, we say it describes a beam of particles with $p_k = \hbar k$ and $E_k = \hbar^2 k^2 / (2m)$. The average density of particles is given by $\rho_k(x, t) = |A|^2$, which is constant. The probability current is

$$\begin{aligned} j_k(x, t) &= -\frac{i\hbar}{2m} \left(\psi_k^* \frac{\partial \psi_k}{\partial x} - \psi_k \frac{\partial \psi_k^*}{\partial x} \right) \\ &= \underbrace{|A|^2}_{\text{density}} \underbrace{\frac{p_k}{m}}_{\text{velocity}} = \text{average flux of particles.} \end{aligned}$$

2.5 Scattering states

Consider a potential barrier, for example, $U(x) = U_0$ for $0 \leq x \leq a$ and $U(x) = 0$ otherwise. A particle travelling towards the barrier can be transmitted and reflected.

Definition 2.1 – Reflection and transmission coefficient

The probability for an incoming particle to be reflected is given by the reflection coefficient

$$R = \lim_{t \rightarrow \infty} \int_{-\infty}^0 |\psi_{\text{GP}}(x, t)|^2 dx.$$

The probability for an incoming particle to be transmitted is given by the transmission coefficient

$$T = \lim_{t \rightarrow \infty} \int_0^{\infty} |\psi_{\text{GP}}(x, t)|^2 dx.$$

Clearly, $R + T = 1$. In the beam interpretation, R and T give respectively the fraction of particles reflected and transmitted. Solving scattering problems using the beam interpretation gives the same results for R and T as the Gaussian wavepacket.

2.5.1 Scattering off a potential step

Consider

$$U(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ U_0 & \text{if } x > 0 \end{cases}$$

where $U_0 \in \mathbb{R}^+$. Recall that the TISE is $-(\hbar^2/(2m))\chi_k'' + U\chi_k = E_k\chi_k$. For $x \leq 0$, we have $\chi_k'' + k^2\chi_k = 0$ where $k = \sqrt{2mE}/\hbar$, so

$$\chi_k(x) = A e^{ikx} + B e^{-ikx}.$$

Here, Ae^{ikx} represents the beam of incident particles and Be^{-ikx} represents the beam of reflected particles. For $x > 0$, we have $\chi''_{\bar{k}} + \bar{k}^2\chi_{\bar{k}} = 0$ where $\bar{k} = \sqrt{2m(E - U_0)}/\hbar$. $\bar{k}$ is real if $E \geq U_0$ and is imaginary if $E < U_0$.

First, consider $E \geq U_0$, then

$$\chi_{\bar{k}}(x) = Ce^{i\bar{k}x} + De^{-i\bar{k}x}.$$

$Ce^{i\bar{k}x}$ is the beam of transmitted particles, and $De^{-i\bar{k}x}$ is an incident beam from $x = \infty$ so we set $D = 0$ due to initial conditions. Combining, we get

$$\chi_{k,\bar{k}}(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } x \leq 0 \\ Ce^{i\bar{k}x} & \text{if } x > 0. \end{cases}$$

Imposing continuity for χ and χ' at $x = 0$, we get

$$B = \frac{k - \bar{k}}{k + \bar{k}}A \text{ and } C = \frac{2k}{k + \bar{k}}A.$$

The probability current is

$$\begin{aligned} J(x, t) &= -\frac{i\hbar}{2m} \left(\chi^* \frac{\partial \chi}{\partial x} - \chi \frac{\partial \chi^*}{\partial x} \right) \\ &= \begin{cases} \frac{\hbar k}{m} (|A|^2 - |B|^2) & \text{if } x \leq 0 \\ \frac{\hbar \bar{k}}{m} |C|^2 & \text{if } x > 0 \end{cases} \end{aligned}$$

so $J_{\text{inc}}(x, t) = \hbar k|A|^2/m$, $J_{\text{ref}}(x, t) = \hbar k|B|^2/m$ and $J_{\text{tra}}(x, t) = \hbar \bar{k}|C|^2/m$. The reflection and transmission coefficients are

$$R = \frac{J_{\text{ref}}}{J_{\text{inc}}} = \left(\frac{k - \bar{k}}{k + \bar{k}} \right)^2 \text{ and } T = \frac{J_{\text{tra}}}{J_{\text{inc}}} = \frac{4k\bar{k}}{(k + \bar{k})^2}.$$

As $E \rightarrow U_0$, $R \rightarrow 1$ and $T \rightarrow 0$. As $E \rightarrow \infty$, $R \rightarrow 0$ and $T \rightarrow 1$.

Now, consider $E < U_0$, then

$$\chi_{\eta}(x) = Ce^{-\eta x} + De^{\eta x}$$

where $\eta = \sqrt{2m(U_0 - E)}/\hbar$. We see that $D = 0$ since otherwise $\chi_{\eta} \rightarrow \infty$ when $x \rightarrow \infty$. Combining, we get

$$\chi_{k,\eta}(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & \text{if } x \leq 0 \\ Ce^{-\eta x} & \text{if } x > 0. \end{cases}$$

Similarly, we impose continuity and get $J_{\text{inc}}(x, t) = \hbar k|A|^2/m$, $J_{\text{ref}}(x, t) = \hbar k|B|^2/m$ and $J_{\text{tra}}(x, t) = 0$. Note that $R = 1$ and $T = 0$ but $\chi_{\eta}(x) \neq 0$ when $x > 0$.

2.5.2 Scattering of a potential barrier

Consider

$$U(x) = \begin{cases} 0 & \text{if } x \leq 0 \text{ or } x \geq a \\ U_0 & \text{if } 0 < x < a \end{cases}$$

where $a, U_0 \in \mathbb{R}^+$. We only consider the case $E < U_0$ here. When $x \leq 0$ or $x \geq a$, we have $k = \sqrt{2mE}/\hbar$. When $0 < x < a$, we have $\eta = \sqrt{2m(U_0 - E)}/\hbar$. So, the solution to the TISE is

$$\chi(x) = \begin{cases} e^{ikx} + Ae^{-ikx} & x \leq 0 \\ Be^{-\eta x} + Ce^{\eta x} & 0 < x < a \\ De^{ikx} + Ee^{-ikx} & x \geq a. \end{cases}$$

$E = 0$ due to the initial conditions.

We have four coefficients with four boundary conditions given by the continuity of χ and χ' at $x = 0$ and

$x = a$.

$$\begin{cases} 1 + A = B + C & \chi \text{ at } x = 0 \\ ik - ikA = -\eta B + \eta C & \chi' \text{ at } x = 0 \\ Be^{-\eta a} + Ce^{\eta a} = De^{ika} & \chi \text{ at } x = a \\ -\eta Be^{-\eta a} + \eta Ce^{\eta a} = ikDe^{ika} & \chi' \text{ at } x = a. \end{cases}$$

Solving this, we get

$$D = \frac{4i\eta k}{(\eta - ik)^2 \exp[(\eta + ik)a] - (\eta + ik)^2 \exp[-(\eta - ik)a]}$$

so

$$T = |D|^2 = \frac{4k^2\eta^2}{(k^2 + \eta^2)^2 \sinh^2(\eta a) + 4k^2\eta^2} \neq 0.$$

This is the quantum tunnelling effect. When $U_0 \gg E$, and so $\eta a \gg 1$, we have

$$T \rightarrow \frac{16k^2\eta^2}{(\eta^2 + k^2)^2} e^{-2\eta a}.$$

Remember this is only the beam interpretation. For a more accurate model, we would have to use the Gaussian wavepacket. After the wavepacket hits the barrier, it splits into two, a reflected and a transmitted wavepacket.

3 Simultaneous measurements in quantum mechanics

In chapter 1, we learnt that Hermitian operators $\hat{O}$ correspond to observables O . They have real eigenvalues, and their eigenstates are orthogonal and form a complete basis. The eigenvalues of $\hat{O}$ are the possible outcomes of a measurement of O . If $\psi = \sum_i a_i \psi_i$ where ψ_i are the eigenstates with eigenvalue λ_i , then $P(O = \lambda_i) = |a_i|^2 = |(\psi_i, \psi)|^2$. After measurement, the wavefunction ψ immediately collapses to ψ_i .

3.1 Commutators

Definition 3.1 – Commutator

The commutator of two operations $\hat{A}$ and $\hat{B}$ is the operator defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

- (i) $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$
- (ii) $[\hat{A}, \hat{A}] = 0$
- (iii) $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$ and $[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$

We can compute $[\hat{x}, \hat{p}]$ in 1D with a test function. Take $\psi \in \mathcal{H}$,

$$\hat{x}\hat{p}\psi = x \left(-i\hbar \frac{\partial}{\partial x} \right) \psi = -i\hbar x \frac{\partial \psi}{\partial x}$$

and

$$\hat{p}\hat{x}\psi = \left(-i\hbar \frac{\partial}{\partial x} \right) (x\psi) = -i\hbar \psi - i\hbar x \frac{\partial \psi}{\partial x}.$$

Hence, $[\hat{x}, \hat{p}]\psi = i\hbar\psi$ so $[\hat{x}, \hat{p}] = i\hbar\hat{I}$ where $\hat{I}$ is the identity. This is the canonical commutator relation.

Definition 3.2 – Simultaneously diagonalisable

Two Hermitian operators $\hat{A}$ and $\hat{B}$ are simultaneously diagonalisable in $\mathcal{H}$ if there exists a complete basis of joint eigenfunctions $\{\psi_i\}$ such that $\hat{A}\psi_i = a_i\psi_i$ and $\hat{B}\psi_i = b_i\psi_i$ where $a_i, b_i \in \mathbb{R}$.

Theorem 3.1

Two Hermitian operators $\hat{A}$ and $\hat{B}$ are simultaneously diagonalisable if and only if $[\hat{A}, \hat{B}] = 0$.

Proof. If $\hat{A}$ and $\hat{B}$ are simultaneously diagonalisable, then there exists a set of joint eigenfunctions $\{\psi_i\}$ that forms a complete basis of $\mathcal{H}$. For all ψ_i , $[\hat{A}, \hat{B}]\psi_i = \hat{A}\hat{B}\psi_i - \hat{B}\hat{A}\psi_i = (a_i b_i - b_i a_i)\psi_i = 0$. Take any $\psi \in \mathcal{H}$, $\psi = \sum_i c_i \psi_i$ for some c_i , so $[\hat{A}, \hat{B}]\psi = \sum_i c_i [\hat{A}, \hat{B}]\psi_i = 0$.

If $[\hat{A}, \hat{B}] = 0$, let ψ_i be an eigenfunction of $\hat{A}$ with eigenvalue λ_i , then $0 = [\hat{A}, \hat{B}]\psi_i = \hat{A}\hat{B}\psi_i - \hat{B}\hat{A}\psi_i = \hat{A}\hat{B}\psi_i - a_i \hat{B}\psi_i$. So, $\hat{A}(\hat{B}\psi_i) = a_i(\hat{B}\psi_i)$, that is $\hat{B}$ maps the eigenspace E_i of $\hat{A}$ with eigenvalue a_i into itself. Hence, $\hat{B}|_{E_i}$ is a Hermitian operator of E_i . Since this holds for all eigenspaces E_i of $\hat{A}$, we can find a complete basis of joint eigenfunctions of $\hat{A}$ and $\hat{B}$. $\square$

3.2 Heisenberg's uncertainty principle

Definition 3.3 – Uncertainty

The uncertainty in a measurement of an observable A on a state ψ is defined as

$$\Delta_\psi A = \sqrt{\langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2}.$$

The expression under the root is equivalent to

$$\begin{aligned} \langle (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})^2 \rangle_\psi &= (\psi, (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})^2 \psi) \\ &= \int_{\mathbb{R}} \psi^* (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})^2 \psi dx \\ &= \underbrace{\int_{\mathbb{R}} \psi^* \hat{A}^2 \psi dx}_{= \langle \hat{A}^2 \rangle_\psi} - 2 \langle \hat{A} \rangle_\psi \underbrace{\int_{\mathbb{R}} \psi^* \hat{A} \psi dx}_{= \langle \hat{A} \rangle_\psi} + \underbrace{\langle \hat{A} \rangle_\psi^2 \int_{\mathbb{R}} \psi^* \psi dx}_{= 1} \\ &= \langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2. \end{aligned}$$

Lemma 3.1

- (i) $\langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2 \geq 0$.
- (ii) $\Delta_\psi A = 0$ if and only if ψ is an eigenfunction of $\hat{A}$.

Proof.

(i)

$$\begin{aligned}\langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2 &= (\psi, (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})^2 \psi) \\ &= ((\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})\psi, (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})\psi) \quad \text{since } A = A^\dagger \\ &= \underbrace{\|(\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})\psi\|}_{\text{call this } \phi}^2 \geq 0\end{aligned}$$

(ii) If $\Delta_\psi A = 0$, then $(\Delta_\psi A)^2 = \|\phi\|^2 = 0$ if and only if $\phi = 0$. So, $\hat{A}\psi = \langle \hat{A} \rangle_\psi \psi$ and hence ψ is an eigenfunction of $\hat{A}$.

If ψ is an eigenfunction of $\hat{A}$ with eigenvalue $a \in \mathbb{R}$, $\langle \hat{A} \rangle_\psi = (\psi, \hat{A}\psi) = a(\psi, \psi) = a$ and $\langle \hat{A}^2 \rangle_\psi = (\psi, \hat{A}^2\psi) = a^2(\psi, \psi) = a^2$. Hence, $(\Delta_\psi A)^2 = \langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2 = 0$.

□

Lemma 3.2

If $\psi, \phi \in \mathcal{H}$, then $|\langle \phi, \psi \rangle|^2 \leq (\phi, \phi)(\psi, \psi)$ with equality if and only if $\phi = a\psi$ with $a \in \mathbb{C}$ and $|a| = 1$.

Proof. By Cauchy-Schwarz inequality.

□

Theorem 3.2 – Generalised uncertainty theorem

If A, B are observables and $\psi \in \mathcal{H}$, then

$$(\Delta_\psi A)(\Delta_\psi B) \geq \frac{1}{2} |(\psi, [\hat{A}, \hat{B}]\psi)|.$$

Proof. $(\Delta_\psi A)^2 = ((\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})\psi, (\hat{A} - \langle \hat{A} \rangle_\psi \hat{I})\psi)$ and similarly for B . Define $\hat{A}' = \hat{A} - \langle \hat{A} \rangle_\psi \hat{I}$ and $\hat{B}' = \hat{B} - \langle \hat{B} \rangle_\psi \hat{I}$ so $(\Delta_\psi A)^2 = (\hat{A}'\psi, \hat{A}'\psi)$ and $(\Delta_\psi B)^2 = (\hat{B}'\psi, \hat{B}'\psi)$. By Lemma 3.2,

$$\begin{aligned}(\Delta_\psi A)^2(\Delta_\psi B)^2 &\geq |(\hat{A}'\psi, \hat{B}'\psi)|^2 \\ &= |(\psi, \hat{A}'\hat{B}'\psi)|^2 \quad \text{since } \hat{A}' \text{ is Hermitian.}\end{aligned}$$

Define the anticommutator $\{\hat{A}', \hat{B}'\} = \hat{A}'\hat{B}' + \hat{B}'\hat{A}'$. Since $\hat{A}', \hat{B}'$ are Hermitian, $[\hat{A}', \hat{B}']^\dagger = -[\hat{A}', \hat{B}']$ and $\{\hat{A}', \hat{B}'\}^\dagger = \{\hat{A}', \hat{B}'\}$. We can write $\hat{A}'\hat{B}' = ([\hat{A}', \hat{B}'] + \{\hat{A}', \hat{B}'\})/2$. Plugging this in,

$$\begin{aligned}(\Delta_\psi A)^2(\Delta_\psi B)^2 &\geq \frac{1}{4} \left| \underbrace{(\psi, [\hat{A}', \hat{B}']\psi)}_{\text{purely imaginary}} + \underbrace{(\psi, \{\hat{A}', \hat{B}'\}\psi)}_{\text{purely real}} \right|^2 \\ &\geq \frac{1}{4} |(\psi, [\hat{A}', \hat{B}']\psi)|^2 + \frac{1}{4} |(\psi, \{\hat{A}', \hat{B}'\}\psi)|^2 \\ &\geq \frac{1}{4} |(\psi, [\hat{A}', \hat{B}']\psi)|^2 \\ &= \frac{1}{4} |(\psi, [\hat{A}, \hat{B}]\psi)|^2 \quad \text{since } [\hat{A}', \hat{B}'] = [\hat{A}, \hat{B}] \text{ as the } \hat{I} \text{ terms cancel.}\end{aligned}$$

□

(i) If $[\hat{A}, \hat{B}] = 0$, then A and B can be measured simultaneously with arbitrary precision on $\psi \in \mathcal{H}$.

(ii) x and p cannot be measured simultaneously to an arbitrary precision since $[\hat{x}, \hat{p}] = i\hbar \hat{I}$ so

$(\Delta_\psi x)(\Delta_\psi p) \geq \hbar/2$ which is Heisenberg's uncertainty principle. We showed that if $\psi = \psi_{\text{GP}}$, $(\Delta_{\psi_{\text{GP}}} x)(\Delta_{\psi_{\text{GP}}} p) = \hbar/2$ at $t = 0$, so this is a state of minimum uncertainty. This can be made more general.

Lemma 3.3

ψ is a state of minimum uncertainty if and only if $\hat{x}\psi = ia\hat{p}\psi$ where $a \in \mathbb{R}$.

Lemma 3.4

The condition for Lemma 3.3 to hold is $\psi = Ce^{-bx^2}$ where $C \in \mathbb{C}$ and $b \in \mathbb{R}^+$.

3.3 Ehrenfest theorem

Theorem 3.3

The expectation value of A evolves according to

$$\frac{d}{dt}\langle \hat{A} \rangle_\psi = \frac{i}{\hbar}\langle [\hat{H}, \hat{A}] \rangle_\psi + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi.$$

Proof.

$$\begin{aligned} \frac{d}{dt}\langle \hat{A} \rangle_\psi &= \frac{d}{dt} \int_{\mathbb{R}} \psi^* \hat{A} \psi dx \\ &= \int_{\mathbb{R}} \frac{\partial}{\partial t} (\psi^* \hat{A} \psi) dx \\ &= \int_{\mathbb{R}} \left[\frac{\partial \psi^*}{\partial t} \hat{A} \psi + \psi^* \frac{\partial \hat{A}}{\partial t} \psi + \psi^* \hat{A} \frac{\partial \psi}{\partial t} \right] dx \\ &= \int_{\mathbb{R}} \left[\frac{\hat{H} \psi^*}{-i\hbar} \hat{A} \psi + \psi^* \frac{\partial \hat{A}}{\partial t} \psi + \psi^* \hat{A} \frac{\hat{H} \psi}{i\hbar} \right] dx \quad \text{by TDSE and (TDSE)*} \\ &= \frac{i}{\hbar} \int_{\mathbb{R}} \psi^* (\hat{H} \hat{A} - \hat{A} \hat{H}) \psi dx + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi \quad \text{by IBP} \\ &= \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle_\psi + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi \end{aligned}$$

□

Example 3.1

(i) Take $\hat{A} = \hat{H}$, then

$$\frac{d}{dt}\langle \hat{H} \rangle_\psi = \frac{dE}{dt} = 0$$

which is the conservation of energy.

(ii) Take $\hat{A} = \hat{p}$, then

$$\frac{d}{dt}\langle \hat{p} \rangle_\psi = -\left\langle \frac{dU}{dx} \right\rangle_\psi$$

where U is the potential. This is the quantum mechanical version of $\dot{p} = -dU/dx$.

(iii) Take $\hat{A} = \hat{x}$, then

$$\frac{d}{dt}\langle \hat{x} \rangle_\psi = \frac{\langle \hat{p} \rangle_\psi}{m}$$

which is what we expect.

3.4 Harmonic oscillator revisited

We have

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where the elastic constant $k = m\omega^2$. To solve the TISE, we want the eigenvalues and eigenfunctions of $\hat{H}$. Rewrite $\hat{H}$ as

$$\hat{H} = \frac{1}{2m}(\hat{p} + im\omega\hat{x})(\hat{p} - im\omega\hat{x}) + \frac{i\omega}{2} \underbrace{[\hat{p}, \hat{x}]}_{= -i\hbar}.$$

We can define

Definition 3.4 – Ladder operators

$$\hat{a} = \frac{1}{\sqrt{2m}}(\hat{p} - im\omega\hat{x}) \text{ and } \hat{a}^\dagger = \frac{1}{\sqrt{2m}}(\hat{p} + im\omega\hat{x})$$

The commutator of the ladder operators is

$$\begin{aligned} [\hat{a}, \hat{a}^\dagger] &= \frac{1}{2m}[\hat{p} - im\omega\hat{x}, \hat{p} + im\omega\hat{x}] \\ &= -\frac{im\omega}{2m}[\hat{x}, \hat{p}] + \frac{im\omega}{2m}[\hat{p}, \hat{x}] \\ &= \hbar\omega\hat{I}. \end{aligned}$$

Also, $[\hat{H}, \hat{a}] = [\hat{a}^\dagger\hat{a}, \hat{a}] = -\hbar\omega\hat{a}$ and $[\hat{H}, \hat{a}^\dagger] = \hbar\omega\hat{a}^\dagger$. Suppose that χ is an eigenfunction of $\hat{H}$ with eigenvalue E , that is $\hat{H}\chi = E\chi$. What is the energy of $\hat{a}\chi$? $\hat{H}(\hat{a}\chi) = \hat{H}\hat{a}\chi = [\hat{H}, \hat{a}]\chi + \hat{a}\hat{H}\chi = -\hbar\omega\hat{a}\chi + E\hat{a}\chi = (E - \hbar\omega)(\hat{a}\chi)$ so $\hat{a}\chi$ is an eigenfunction of $\hat{H}$ with eigenvalue $E - \hbar\omega$. Analogously, $\hat{a}^\dagger\chi$ is an eigenfunction of $\hat{H}$ with eigenvalue $E + \hbar\omega$. We can easily prove by induction that $\hat{a}^n\chi$ is an eigenfunction of $\hat{H}$ with eigenvalue $E - n\hbar\omega$, and $(\hat{a}^\dagger)^n\chi$ is an eigenfunction of $\hat{H}$ with eigenvalue $E + n\hbar\omega$. Using the fact that $\langle\hat{H}\rangle \geq 0$, there cannot be eigenfunctions of arbitrarily low energy. Hence, there exists an eigenfunction χ_0 such that $\hat{a}\chi_0 = 0$. So,

$$\begin{aligned} \frac{1}{\sqrt{2m}}(\hat{p} - im\omega\hat{x})\chi_0 &= 0 \\ -i\hbar\frac{d\chi_0}{dx} - im\omega x\chi_0 &= 0 \\ \chi_0(x) &= Ce^{-m\omega x^2/2\hbar}. \end{aligned}$$

Also,

$$\hat{H}\chi_0 = \hat{a}^\dagger \underbrace{\hat{a}\chi_0}_{=0} + \frac{\hbar\omega}{2}\hat{I}\chi_0 = \frac{\hbar\omega}{2}\chi_0$$

so $E_0 = \hbar\omega/2$. To find states with $E > E_0$,

$$\chi_n = (\hat{a}^\dagger)^n\chi_0 = \frac{C}{(2m)^{n/2}} \left(-i\hbar\frac{d}{dx} + im\omega x \right)^n e^{-m\omega x^2/2\hbar}$$

corresponding to $E_n = (n + 1/2)\hbar\omega$. We can also find the Hermite polynomials this way.

4 3D solutions of the Schrödinger equation

The TISE in 3D reads

$$-\frac{\hbar^2}{2m}\nabla^2\chi(\mathbf{x}) + U(\mathbf{x})\chi(\mathbf{x}) = E\chi(\mathbf{x}).$$

We can write the Laplacian in different coordinate systems. In Cartesian coordinates,

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

In spherical coordinates (r, θ, φ) where θ is the angle with the z -axis so $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$,

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} (r^2) + \frac{1}{r^2 \sin^2 \theta} \left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{\partial^2}{\partial \varphi^2} \right].$$

In this course, we will always deal with spherically symmetric potentials $U(\mathbf{x}) = U(r)$. Even if U is spherically symmetric, the eigenfunctions of $\hat{H}$ will also depend on θ and φ .

4.1 Angular momentum

In classical mechanics, $\mathbf{L} = \mathbf{x} \times \mathbf{p}$. When we have a spherically symmetric potential, then

$$\frac{d\mathbf{L}}{dt} = \dot{\mathbf{x}} \times \mathbf{p} + \mathbf{x} \times \dot{\mathbf{p}} = \mathbf{0}.$$

This implies that the problem can be treated as a 2D problem by taking the plane $\mathbf{L} \cdot \mathbf{x} = 0$. Then, we can just solve in the $\mathbf{e}_r$ direction and it becomes a 1D problem.

Definition 4.1 – Angular momentum operator

The angular momentum operator is

$$\hat{\mathbf{L}} = \hat{\mathbf{x}} \times \hat{\mathbf{p}} = -i\hbar \mathbf{x} \times \nabla$$

in 3D where $\hat{\mathbf{x}} = \mathbf{x} = (x, y, z)$ and $\hat{\mathbf{p}} = -i\hbar \nabla = (-i\hbar \partial/\partial x, -i\hbar \partial/\partial y, -i\hbar \partial/\partial z)$. In Cartesian coordinates,

$$\hat{L}_i = -i\hbar \epsilon_{ijk} x_j \frac{\partial}{\partial x_k} = \epsilon_{ijk} \hat{x}_j \hat{p}_k.$$

$\hat{L}_i$ is Hermitian. Also, $[\hat{L}_i, \hat{L}_j] \neq 0$ if $i \neq j$, so different components of $\mathbf{L}$ cannot be determined simultaneously. In fact, $[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k$ since

$$\begin{aligned} & [\hat{L}_1, \hat{L}_2]\chi(x_1, x_2, x_3) \\ &= -\hbar^2 \left[\left(x_2 \frac{\partial}{\partial x_3} - x_3 \frac{\partial}{\partial x_2} \right) \left(x_3 \frac{\partial}{\partial x_1} - x_1 \frac{\partial}{\partial x_3} \right) - \left(x_3 \frac{\partial}{\partial x_1} - x_1 \frac{\partial}{\partial x_3} \right) \left(x_2 \frac{\partial}{\partial x_3} - x_3 \frac{\partial}{\partial x_2} \right) \right] \chi \\ &= -\hbar^2 \left(x_2 \frac{\partial}{\partial x_1} - x_1 \frac{\partial}{\partial x_2} \right) \chi \\ &= i\hbar \hat{L}_3 \chi. \end{aligned}$$

Similarly for others.

Definition 4.2 – Total angular momentum operator

The total angular momentum operator is

$$\hat{L}^2 = \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2.$$

We can prove that $[\hat{L}^2, \hat{L}_i] = 0$ so we can find a joint set of eigenfunctions of $\hat{L}^2$ and one of the components. By convention, we take $\hat{L}_3$ which corresponds with the z-axis. To find the joint set of eigenfunctions of $\hat{L}^2$ and $\hat{L}_3$, write $\hat{L}$ in spherical coordinates and get

$$\hat{L}_3 = -i\hbar \frac{\partial}{\partial \varphi} \text{ and } \hat{L}^2 = -\frac{\hbar^2}{\sin^2 \theta} \left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{\partial^2}{\partial \varphi^2} \right].$$

The joint eigenfunctions depend on θ and φ so

$$\begin{cases} -i\hbar \frac{\partial}{\partial \varphi} Y(\theta, \varphi) = \hbar m Y(\theta, \varphi) \\ -\frac{\hbar^2}{\sin^2 \theta} \left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{\partial^2}{\partial \varphi^2} \right] Y(\theta, \varphi) = \lambda Y(\theta, \varphi) \end{cases}$$

where m is just a variable and not mass. We try separation of variables $Y(\theta, \varphi) = y(\theta)X(\varphi)$ and we get

$$\begin{aligned} -i\hbar \frac{\partial X(\varphi)}{\partial \varphi} y(\theta) &= \hbar m X(\varphi) y(\theta) \\ X(\varphi) &= e^{im\varphi}. \end{aligned}$$

Given that the wavefunction is single-valued in $\mathbb{R}^3$, $X(\varphi)$ must be invariant under $\varphi \mapsto \varphi + 2\pi$, so $e^{im(2\pi)} = 1$ and $m \in \mathbb{Z}$. From the second equation, we get

$$-\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial y(\theta)}{\partial \theta} \right) - \frac{m^2}{\sin^2 \theta} y(\theta) = -\frac{\lambda}{\hbar^2} Y(\theta).$$

This is the associate Legendre equation and its solution is given by

$$y(\theta) = P_{l,m}(\cos \theta) = \sin^{|m|} \theta \frac{d^{|m|}}{d(\cos \theta)^{|m|}} P_l(\cos \theta)$$

where P_l are the ordinary Legendre polynomials. Since P_l is a polynomial of degree l , we must have $-l \leq m \leq l$. Without proof, the eigenvalues of $\hat{L}^2$ are given by $\lambda = \hbar^2 l(l+1)$. Putting everything together, $Y_{l,m}(\theta, \varphi) = P_{l,m}(\cos \theta) e^{im\varphi}$ with $l \in \mathbb{Z}_{\geq 0}$ and $m \in \{-l, -l+1, \dots, l\}$ are called the spherical harmonics. These satisfy

$$\begin{cases} \hat{L}^2 Y_{l,m}(\theta, \varphi) = \hbar^2 l(l+1) Y_{l,m}(\theta, \varphi) \\ \hat{L}_3 Y_{l,m}(\theta, \varphi) = \hbar m Y_{l,m}(\theta, \varphi). \end{cases}$$

l and m are called quantum numbers. l characterises the total angular momentum and m is the azimuthal number that characterises the z-component of angular momentum. In classical mechanics, $-|\mathbf{L}| \leq L_z \leq |\mathbf{L}|$. This is similar to the quantum mechanical version $-l \leq m \leq l$. The first few spherical harmonics are

$$\begin{aligned} Y_{0,0}(\theta, \varphi) &= 1/\sqrt{4\pi} \\ Y_{1,0}(\theta, \varphi) &= \sqrt{3/(4\pi)} \cos \theta \\ Y_{1,\pm 1}(\theta, \varphi) &= \mp \sqrt{3/(8\pi)} \sin \theta e^{\pm i\varphi}. \end{aligned}$$

Like for all eigenfunctions of a Hermitian operator, we have $(Y_{l,m}, Y_{l',m'}) = \delta_{ll'} \delta_{mm'}$.

4.2 3D TISE for spherically symmetric potentials

In spherical coordinates,

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} (r^2) + \frac{1}{r^2 \sin^2 \theta} \left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{\partial^2}{\partial \varphi^2} \right].$$

Looking at the definition of $\hat{L}^2$ in spherical coordinates, we see that

$$-\hbar^2 \nabla^2 = -\frac{\hbar^2}{r} \frac{\partial^2}{\partial r^2} (r^2) + \frac{\hat{L}^2}{r^2}$$

so the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2mr} \frac{\partial^2}{\partial r^2} (r^2) + \frac{\hat{L}^2}{2mr^2} + U(\mathbf{x}).$$

For a spherically symmetric potential, $U(\mathbf{x}) = U(r)$. For this kind of potential, $[\hat{H}, \hat{L}^2] = 0$ and $[\hat{H}, \hat{L}_i] = 0$. To prove this, we need to look at

$$[\hat{L}_i, \hat{x}_j] = [\varepsilon_{imn} \hat{x}_m \hat{p}_n, \hat{x}_j] = \varepsilon_{imn} (\hat{x}_m [\hat{p}_n, \hat{x}_j] + [\hat{x}_m, \hat{x}_j] \hat{p}_n).$$

The canonical commutator relation is $[\hat{p}_n, \hat{x}_j] = -i\hbar \delta_{nj}$ in 3D, so

$$\begin{aligned} [\hat{L}_i, \hat{x}_j] &= \varepsilon_{imn} \hat{x}_m (-i\hbar \delta_{nj}) \\ &= -i\hbar \varepsilon_{imj} \hat{x}_m \\ &= i\hbar \varepsilon_{ijm} \hat{x}_m. \end{aligned}$$

Also,

$$\begin{aligned} [\hat{L}_i, \hat{x}_j^2] &= [\hat{L}_i, \hat{x}_j] \hat{x}_j + \hat{x}_j [\hat{L}_i, \hat{x}_j] \\ &= i\hbar \varepsilon_{ijm} (\hat{x}_m \hat{x}_j + \hat{x}_j \hat{x}_m) \\ &= 0 \end{aligned}$$

since it is the product of something symmetric and something antisymmetric. Moreover, $[\hat{L}_i, U(r)] = 0$ by substituting $r = (\hat{x}_1^2 + \hat{x}_2^2 + \hat{x}_3^2)^{1/2}$. Analogously, we have $[\hat{L}_i, \hat{p}_j] = i\varepsilon_{ijm} \hat{p}_m$ and $[\hat{L}_i, \hat{p}_j^2] = 0$ so $[\hat{L}_i, \hat{p}^2] = 0$. This gives the result $[\hat{H}, \hat{L}^2] = 0$ and $[\hat{H}, \hat{L}_i] = 0$.

Hence, $\{\hat{H}, \hat{L}^2, \hat{L}_3\}$ are a set of mutually commuting operators.

- (i) We can find a joint set of eigenstates of these three operators that form a basis of $\mathcal{H}$.
- (ii) The eigenvalues of these operators (which are $\{E, |\mathbf{L}|^2, L_z\}$) can be simultaneously measured to an arbitrary precision.
- (iii) The set of operators is maximal. We cannot construct another independent operator (other than $\hat{I}$) that commutes with them.

Let us start looking for solutions of the TISE $\chi(r)$ such that $\hat{L}^2 \chi(r) = 0$. We have

$$\nabla^2 \chi(r) = \frac{1}{r} \frac{\partial^2}{\partial r^2} (r \chi(r))$$

so the TISE reads

$$-\frac{\hbar^2}{2m} \left(\frac{d^2 \chi}{dr^2} + \frac{2}{r} \frac{d\chi}{dr} \right) + U(r) \chi(r) = E \chi(r).$$

The normalisation condition is

$$\int_{\mathbb{R}^3} |\chi(r)|^2 dV = 4\pi \int_0^\infty |\chi(r)|^2 r^2 dr = 1.$$

We need the eigenfunction $\chi(r) \rightarrow 0$ sufficiently fast. We also need $\chi(r)$ to behave well at $r \rightarrow 0$. The most singular behaviour allowed is $\sim 1/r$.

Now, how do we actually solve the TISE? Define $\sigma(r) = r\chi(r)$ so

$$-\frac{\hbar^2}{2m} \frac{d^2\sigma(r)}{r^2} + U(r)\sigma(r) = E\sigma(r).$$

This is the 1D TISE defined only on $\mathbb{R}^+$ with the normalisation condition

$$\int_0^\infty |\sigma(r)|^2 dr = \frac{1}{4\pi}.$$

We want $\sigma(r) = 0$ and $\sigma'(r)$ to be finite at $r = 0$. We can solve for $\sigma(r)$ on $\mathbb{R}$ and then only look for odd solutions.

The normalisation condition would be satisfied if $\sigma(r) \sim a$ as $r \rightarrow 0$, so why do we require that $\sigma(r) = 0$ at $r = 0$? If we allowed $\sigma(r) \sim a$ and $\chi(r) \sim a/r$ as $r \rightarrow 0$, then $\hat{H}$ would not be Hermitian.

Proof. For $\hat{H}$ to be Hermitian, we want $(\phi, \hat{H}\chi) = (\hat{H}\phi, \chi)$ for all $\phi, \chi \in \mathcal{H}$. Here, we have bound states so we can make $\phi, \chi \in \mathbb{R}$ by example sheet 1.

$$\begin{aligned} (\phi, \hat{H}\chi) &= \int_0^\infty \phi(r) \hat{H}\chi(r) r^2 dr \\ &= -\frac{\hbar^2}{2m} \int_0^\infty \phi(r) \frac{d}{dr} \left(r^2 \frac{d\chi}{dr} \right) dr \\ &= -\frac{\hbar^2}{2m} \left[r^2 \phi \frac{d\chi}{dr} - r^2 \chi \frac{d\phi}{dr} \right]_0^\infty - \underbrace{\frac{\hbar^2}{2m} \int_0^\infty \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) \chi(r) dr}_{= (\hat{H}\phi, \chi)}. \end{aligned}$$

If $\phi(r) \sim B$ and $\chi(r) \sim A/r$ as $r \rightarrow 0$, then $r^2\phi(d\chi/dr) - r^2\chi(d\phi/dr) \not\rightarrow 0$ as $r \rightarrow 0$. Hence, $(\phi, \hat{H}\chi) \neq (\hat{H}\phi, \chi)$ so $\hat{H}$ is not Hermitian. From the quantum mechanical interpretation, $\hat{H}$ is Hermitian, so we classify $\chi(r) \sim A/r$ as unphysical. $\square$

Example 4.1

Consider the 3D spherically symmetric finite potential well

$$U(r) = \begin{cases} 0 & \text{if } r \leq a \\ U_0 & \text{if } r > a \end{cases}$$

with $a \in \mathbb{R}^+$ and $U_0 \in \mathbb{R}^+$. We extend the potential to $r \in \mathbb{R}$ by

$$U(r) = \begin{cases} 0 & \text{if } |r| \leq a \\ U_0 & \text{if } |r| > a. \end{cases}$$

We only consider $l = 0$ here and solve for $\sigma(r) = r\chi(r)$ on $\mathbb{R}$. We look for bound states with $0 \leq E \leq U_0$. The TISE reads

$$-\frac{\hbar^2}{2m} \frac{d^2\sigma}{dr^2} + U\sigma = E\sigma$$

and the solution is

$$\sigma(r) = \begin{cases} A \sin(kr) + B \cos(kr) & \text{if } |r| \leq a \\ C e^{-\bar{k}r} & \text{if } r > a \\ D e^{\bar{k}r} & \text{if } r < -a \end{cases}$$

where $k = \sqrt{2mE}/\hbar$ and $\bar{k} = \sqrt{2m(U_0 - E)}/\hbar$. We only want odd solutions so $B = 0$ and $D = -C$. The

boundary condition at $r = a$ gives

$$\begin{cases} A \sin(ka) = Ce^{-\bar{k}a} \\ kA \cos(ka) = -\bar{k}Ce^{-\bar{k}a} \end{cases}$$

by the continuity of σ and σ' . Taking their ratios and from the definitions of k and k' , we have

$$\begin{cases} -k \cot(ka) = \bar{k} \\ k^2 + \bar{k}^2 = 2mU_0/\hbar^2. \end{cases}$$

Plotting a graph of $\bar{k}$ against k , we see that

$$U_0 \geq \frac{\pi^2 \hbar^2}{8ma^2}$$

is required for a solution to exist. There is not always a bound state unlike the 1D case. We need U_0 to be above a certain threshold.

$$\chi(r) = \begin{cases} A \sin(kr)/r & \text{if } r \leq a \\ Ce^{-\bar{k}r}/r & \text{if } r > a. \end{cases}$$

4.3 The hydrogen atom

We model the nucleus to be stationary at the origin, that is $m_p \gg m_e$. We use the Coulomb force,

$$F(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r^2} \text{ and } U(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}.$$

We only look for bound states with $E < 0$. The TISE is

$$-\frac{\hbar^2}{2m_e} \nabla^2 \chi(r, \theta, \varphi) - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} \chi(r, \theta, \varphi) = E \chi(r, \theta, \varphi)$$

but

$$-\hbar^2 \nabla^2 = -\frac{\hbar^2}{r} \frac{\partial^2}{\partial r^2} (r \) + \frac{\hat{L}^2}{r^2}$$

so

$$-\frac{\hbar^2}{2m_e} \frac{1}{r} \frac{\partial^2}{\partial r^2} (r \chi) + \frac{\hat{L}^2}{2m_e r^2} \chi - \frac{e^2}{4\pi\epsilon_0 r} \chi = E \chi.$$

Since χ must be an eigenfunction of $\hat{L}^2$ and $\hat{L}_3$, we can separate variables and look for solutions of the form $\chi(r, \theta, \varphi) = R(r)Y_{l,m}(\theta, \varphi)$. So,

$$\hat{L}^2 R(r)Y_{l,m}(\theta, \varphi) = \hbar^2 l(l+1)R(r)Y_{l,m}(\theta, \varphi).$$

Plugging this in, we get

$$\begin{aligned} -\frac{\hbar^2}{2m_e} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) Y_{l,m} + \frac{\hbar^2}{2m_e r^2} l(l+1) R Y_{l,m} - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} R Y_{l,m} &= E R Y_{l,m} \\ -\frac{\hbar^2}{2m_e} \left(\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} \right) + \underbrace{\left(-\frac{e^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2 l(l+1)}{2m_e r^2} \right)}_{\text{the effective potential } U_{\text{eff}}(r)} R &= E R \end{aligned}$$

4.3.1 The case $l = 0$

We have $U_{\text{eff}}(r) = U(r)$. Let $v^2 = -2m_e E / \hbar^2 > 0$ and $\beta = e^2 m_e / (2\pi\epsilon_0 \hbar^2)$, so the equation becomes

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \left(\frac{\beta}{r} - v^2 \right) R = 0.$$

The asymptotic behaviour as $r \rightarrow \infty$ is determined by

$$\frac{d^2 R}{dr^2} - v^2 R = 0 \implies R(r) \sim e^{\pm vr} \text{ as } r \rightarrow \infty.$$

By normalisation, we pick the negative sign in the exponent so $R(r) \sim e^{-vr}$. Also, at $r = 0$, we need R to be finite.

Exploiting the asymptotic behaviour, try the ansatz $R(r) = f(r)e^{-vr}$ to get

$$f''(r) + \frac{2}{r}(1 - vr)f'(r) + \frac{1}{r}(\beta - 2v)f(r) = 0.$$

This is a homogeneous ODE with a regular point at $r = 0$, so we can write

$$f(r) = \sum_{n=0}^{\infty} a_n r^{c+n}.$$

Its derivatives are

$$f'(r) = \sum_{n=0}^{\infty} a_n (c+n) r^{c+n-1} \text{ and } f''(r) = \sum_{n=0}^{\infty} a_n (c+n)(c+n-1) r^{c+n-2}.$$

Substituting in, we get

$$\sum_{n=0}^{\infty} \left[a_n (c+n)(c+n-1) r^{c+n-2} + 2a_n (1 - vr)(c+n) r^{c+n-2} + a_n (\beta - 2v) r^{c+n-1} \right] = 0.$$

From the lowest power of r , which is r^{c-2} , we find that

$$a_0 c(c-1) + 2a_0 c = 0 \implies c = -1 \text{ or } 0.$$

$c = -1$ would give $\chi \sim A/r$ near $r = 0$, which violates normalisation, so we pick $c = 0$. This gives

$$\begin{aligned} \sum_{n=1}^{\infty} [a_n n(n+1) + a_{n-1}(\beta - 2vn)] r^{n-2} &= 0 \\ \implies a_n &= \frac{2vn - \beta}{n(n+1)} a_{n-1} \end{aligned}$$

Proposition 4.1

If the series $f(r) = \sum_{n=0}^{\infty} a_n r^n$ does not terminate, then $R(r)$ would not be normalisable.

Proof. As $n \rightarrow \infty$, $a_n / a_{n-1} \rightarrow 2v/n$. This is the same asymptotic behaviour as $g(r) = e^{2vr} = \sum_{n=0}^{\infty} (2v)^n / n! r^n$. Hence, $f(r) \sim e^{2vr}$ and $R(r) \sim e^{vr}$, which is not normalisable. $\square$

The series must terminate, so there exists $N > 0$ such that $a_N = 0$ with $a_{N-1} \neq 0$. So,

$$\frac{2vN - \beta}{N(N+1)} a_{N-1} = 0 \implies v = \frac{\beta}{2N}.$$

This gives

$$E_N = -\frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{N^2}$$

with $N \in \mathbb{Z}_{\geq 1}$. To solve for the eigenfunctions $R_N(r)$, notice that

$$a_n = -2\nu \frac{N-n}{n(n+1)} a_{n-1}.$$

We can use this to find the coefficients of the polynomial $R_N(r)$, which is of degree $N-1$. Setting $a_0 = 1$, we get

$$\begin{aligned} R_1(r) &= A_1 e^{-\nu r} \\ R_2(r) &= A_2 (1 - \nu r) e^{-\nu r} \\ R_3(r) &= A_3 (1 - 2\nu r + (2\nu^2/3)r^2) e^{-\nu r} \end{aligned}$$

where A_i are the normalisation constants. In general, $R_N(r) = L_N(\nu r) e^{-\nu r}$ where L_N are the Laguerre polynomials. We get the wavefunctions via $\chi_{N,0,0} = R_N Y_{0,0}$. We can get that $\langle \hat{r} \rangle_{\chi_{1,0,0}} = (3/2)a_0$ where a_0 is the Bohr radius. In fact, the Bohr radius is the radius where the probability density $P(r) \propto r^2 |R_1(r)|^2$ is at maximum.

4.3.2 The case $l > 0$

In terms of ν and β , we have

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \left(\frac{\beta}{r} - 2\nu - \frac{l(l+1)}{r^2} \right) R = 0.$$

The asymptotic behaviour $R(r) \sim e^{-\nu r}$ as $r \rightarrow \infty$ still holds, so we try $R(r) = g(r) e^{-\nu r}$.

$$g'' + \frac{2}{r} (1 - \nu r) g' + \left(\frac{\beta}{r} - 2\nu - \frac{l(l+1)}{r^2} \right) g = 0$$

and we express $g(r)$ as the power series $\sum_{n=0}^{\infty} a_n r^{\sigma+n}$. Looking at the lowest power of r , we have

$$a_0 [\sigma(\sigma-1) + 2\sigma - l(l+1)] = 0 \implies \sigma = -l-1 \text{ or } l.$$

$\sigma = -l-1$ violates normalisation so we choose $\sigma = l$. Hence, $g(r) = \sum_{n=0}^{\infty} a_n r^{l+n}$ where

$$a_n = \frac{2\nu(n+l) + \beta}{n(n+2l-1)} a_{n-1}.$$

As before, there exists $n_{\max} > 0$ such that $a_{n_{\max}} = 0$ with $a_{n_{\max}-1} \neq 0$, so $2\nu(n_{\max} + l) - \beta = 0$. Setting $N = n_{\max} + l$, we get $\nu = \beta/(2N)$. Substituting in the definition of ν , we have

$$E_N = -\frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{N^2}$$

as before. The eigenvalues of $\hat{H}$ are the same but there is more degeneracy. We can have $l \in \{0, 1, \dots, N-1\}$ and $m \in \{-l, -l+1, \dots, l\}$ so

$$\text{the number of eigenfunctions with energy } E_N = \sum_{l=0}^{N-1} \sum_{m=-l}^l = N^2.$$

In summary,

$$\chi_{N,l,m}(r, \theta, \varphi) = R_{N,l}(r) Y_{l,m}(\theta, \varphi) = r^l g_{N,l}(r) e^{-\beta r/(2N)} Y_{l,m}(\theta, \varphi)$$

where $g_{N,l}$ is a polynomial of degree $N-l-1$ defined by

$$g_{N,l}(r) = \sum_{k=0}^{N-l-1} a_k^{(N,l)} r^k \text{ with } a_k^{(N,l)} = \frac{2\nu}{k} \frac{k+l-N}{k+2l+1} a_{k-1}^{(N,l)}.$$

These polynomials are called the generalised Laguerre polynomials. There are three quantum numbers:

$N = 1, 2, 3 \dots$	principal quantum number
$l = 0, 1, \dots, N - 1$	total angular momentum quantum number
$m = -l, -l + 1, \dots, l$	azimuthal quantum number

We can compute the wavefunctions for different orbitals of the hydrogen atom:

1s	$a_0^{(1,0)} e^{-\beta r/2}$
2s	$(a_0^{(2,0)} + a_1^{(2,0)} r) e^{-\beta r/4}$
2p	$r a_0^{(2,1)} e^{\beta r/4} \cos \theta$ $r a_0^{(2,1)} e^{-\beta r/4} \sin \theta e^{\pm i\varphi}$
3s	$(a_0^{(3,0)} + a_1^{(3,0)} r + a_2^{(3,0)} r^2) e^{-\beta r/6}$
3p	$r(a_0^{(3,1)} + a_1^{(3,1)} r) e^{-\beta r/6} \cos \theta$ $r(a_0^{(3,1)} + a_1^{(3,1)} r) e^{-\beta r/6} \sin \theta e^{\pm i\varphi}$

These still require normalisation.

We can also try to generalise the hydrogen atom. Treat the Z electrons as independent (which is absolutely not true) and write $\chi(\mathbf{x}_1, \dots, \mathbf{x}_Z) = \chi(\mathbf{x}_1) \dots \chi(\mathbf{x}_Z)$. We will get $E = \sum_{i=1}^Z E_i$. This is a horrible approximation.